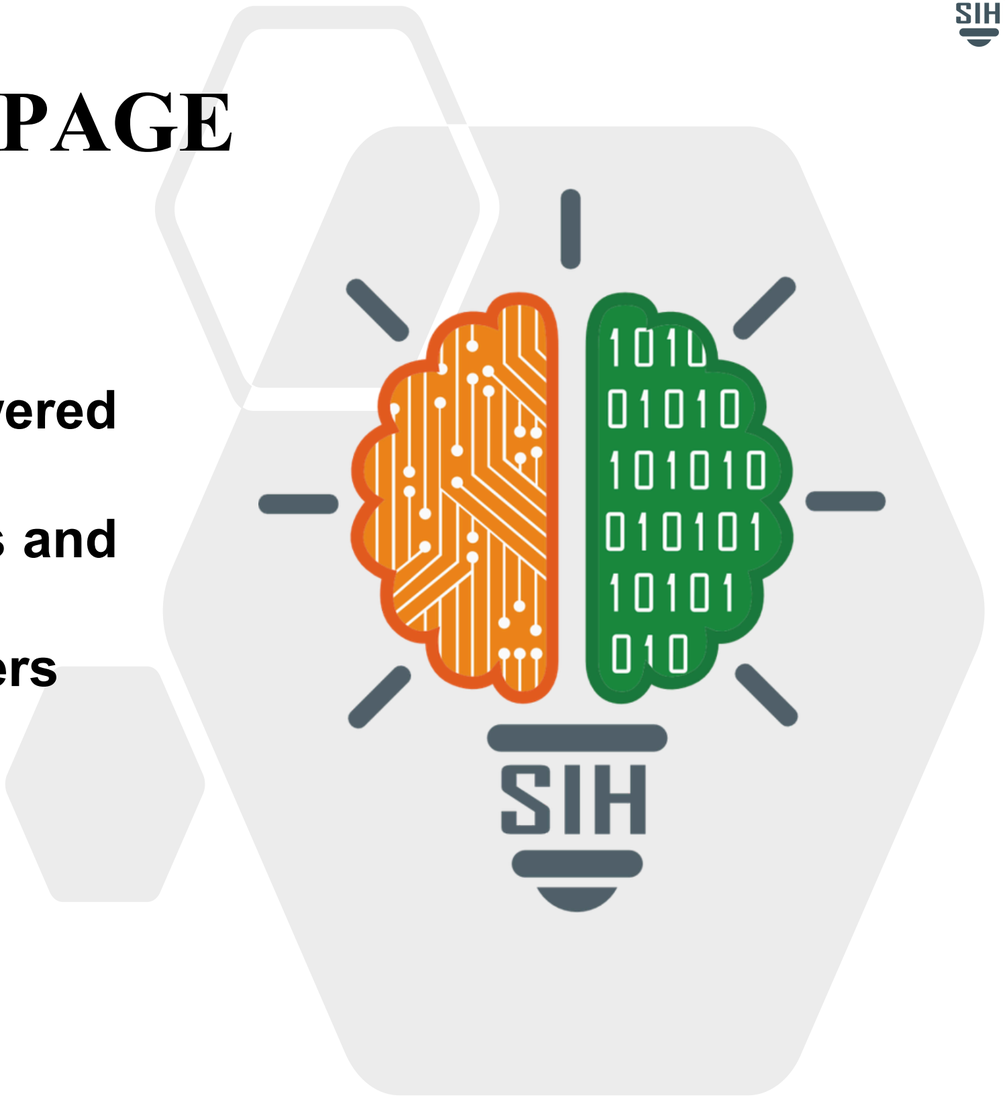
SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID – SIH26107**
- **Problem Statement Title - AI-powered Intelligent Assistant for Indian Standards and BIS Services for Industries and Consumers**
- **Theme - Smart Automation**
- **PS Category- Software**
- **Team ID - 140981**
- **Team Name - SLUG**



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IDEA TITLE

'mation Access C

PROBLEM

- Scattered Information: BIS standards, certification schemes, licensing, testing and consumer services are spread across documents and portals.
- Expertise Barrier: Users often struggle to identify the right standard, scheme or certification requirement for a product.
- Time-Consuming Search: Finding the applicable clause, procedure or testing requirement manually through PDFs is inefficient.
- Low Accessibility: Information is difficult to consume for MSMEs, startups, students and consumers, especially across languages.

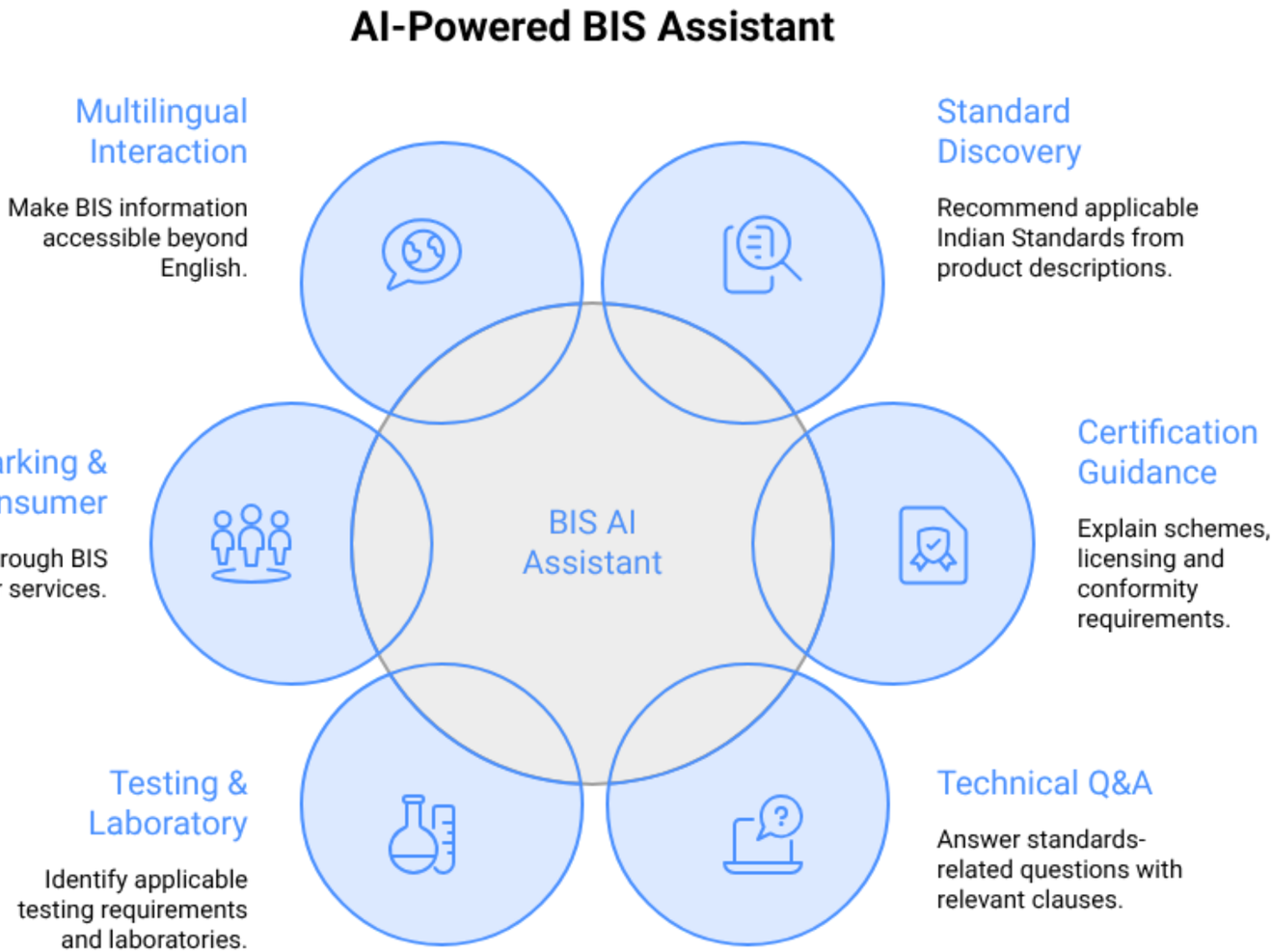
IDEA

Make BIS knowledge conversational.

From searching documents → to asking the right question.

SOLUTION

- One assistant for the entire BIS information journey.
- Standard Discovery → Recommend applicable Indian Standards from product descriptions.
- Certification Guidance → Explain schemes, licensing and conformity requirements.
- Technical Q&A → Answer standards-related questions with relevant clauses.
- Testing & Laboratory Support → Identify applicable testing requirements and laboratories.
- Hallmarking & Consumer Support → Guide users through BIS consumer services.
- Multilingual Interaction → Make BIS information accessible beyond English.
- **Every answer is source-backed with relevant BIS documents or clauses.**



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TECHNICAL APPROACH

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Technical foundation:

Frontend: React + TypeScript + Vite

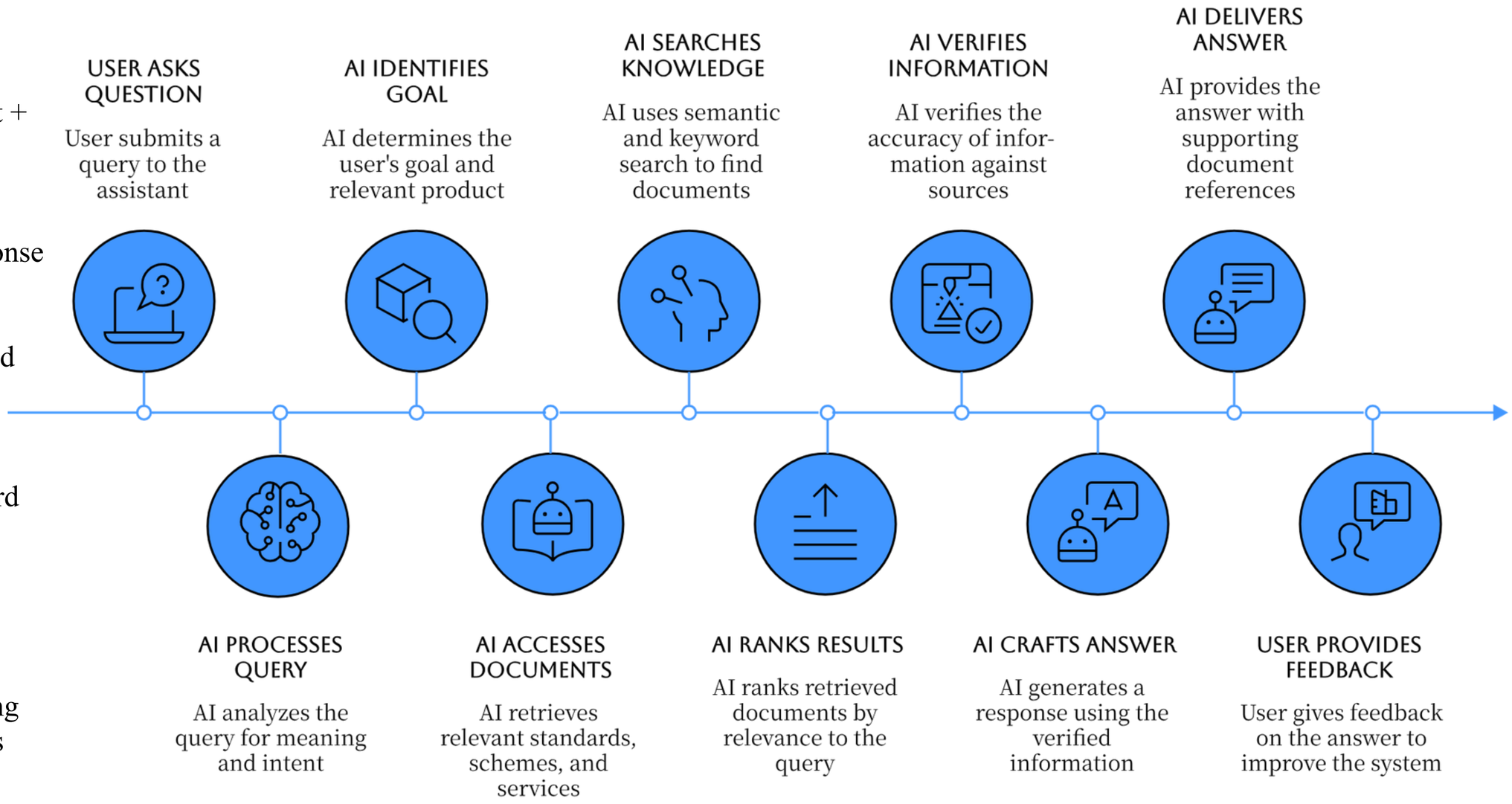
AI Layer: NLP/LLM-based query understanding and response generation

Knowledge Layer: Authorized BIS documents, standards and service information

Retrieval: Semantic + keyword search

Grounding: Relevant document/section/clauses attached to answers

Backend: API layer connecting users, retrieval and AI services



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FEASIBILITY AND VIABILITY

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Feasibility — HIGH

- Uses established LLM, RAG and semantic-search technologies.
- Can be built around authorized BIS knowledge sources.
- Modular architecture allows individual services to be expanded independently.
- Existing frontend foundation uses React, TypeScript and Vite.

Key Risks

- Incorrect or outdated standards information
- Hallucinated answers
- Ambiguous product descriptions
- Large and complex standards documents
- Multilingual accuracy

Mitigation

- Source-grounded retrieval
- Clause-level citations
- Confidence/relevance thresholds
- Document version control
- Human escalation for uncertain cases

Feasibility

Can we build it?

Proven Technologies
- Uses established LLM, RAG, NLP and search technologies

Authorized BIS Sources
- Works with standards, schemes and BIS service documents

Modular Architecture
- Enables phased rollout of features and services

Source-Backed Answers
- Every reply links to the document or clause

Multilingual Support
- Uses proven translation and multilingual language models

Challenges & Risks

What can go wrong?

Scattered Data
- Standards, schemes and PDFs in varied formats

Incorrect Answers
- Similar standards or clauses retrieved by mistake

AI Hallucination Risk
- Unsupported or fabricated responses, low confidence

Outdated Information
- Superseded standards and changing schemes

Traceability & Trust
- Need for citations and explainability

Scalability & Languages
- High query volume across many Indian languages

Our Strategies

How do we solve it?

Document Processing
- Parse and index standards, PDFs and web content

Hybrid Retrieval (RAG)
- Semantic + Keyword + Metadata search

Confidence Scoring & Validation
- Answer only when grounded in authorized sources

Regular Knowledge Updates
- Sync with latest BIS publications and schemes

Clause-Level Citations
- Link answers to source documents and clauses

Modular & Scalable Architecture
- Independent services and multilingual scalability

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IMPACT AND BENEFITS

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For Consumers & Students

- Plain-language answers on standards & BIS
- Single window for BIS schemes and services
- Guidance on hallmarking & consumer queries
- Multilingual access for citizens across India

For MSMEs & Industry

- Faster identification of applicable standards
- Clear certification and licensing guidance
- Less time searching across PDFs and portals
- Suggestions for relevant testing laboratories

For BIS & Technology

- Retrieval from authorized BIS sources
- Clause-level, source-backed responses
- Context-aware product-to-standard mapping
- Multilingual conversational AI agent

Strategic Impact

Scattered Information → Natural Language Query → Source-Backed Guidance → Faster Compliance & Trust

Long-Term Vision

ONE NATION
ONE INTELLIGENT ASSISTANT
ONE TRUSTED GATEWAY TO BIS

Key Differentiators

AI-assisted	Source-backed	Accurate	Multilingual	Accessible
Conversational understanding	References to documents & clauses	Answers from authorized BIS knowledge sources	Support for regional language interaction	Built for MSMEs, students & consumers

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RESEARCH AND REFERENCES

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- **BIS — Indian Standards & Standards Information**
  Primary source for accessing Indian Standards, technical information and standards-related services.
- **BIS — Product Certification Schemes**
  Relevant to understanding licensing, conformity assessment and certification requirements.
- **BIS — Hallmarking & Consumer Services**
  Supports information related to hallmarking, consumer queries and BIS consumer services.
- **BIS — Laboratory Recognition Scheme**
  Relevant to identifying recognized laboratories and guiding users toward applicable testing facilities.
- **ISO/IEC 42001:2023 — Artificial Intelligence Management System**
  Provides a framework for responsible management and governance of AI systems used in organizational applications.